

Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing

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SUMMARY

The common approach to the multiplicity problem calls for controlling the familywise error rate (FWER). This approach, though, has faults, and we point out a few. A different approach to problems of multiple significance testing is presented. It calls for controlling the expected proportion of falsely rejected hypotheses—the false discovery rate. This error rate is equivalent to the FWER when all hypotheses are true but is smaller otherwise. Therefore, in problems where the control of the false discovery rate rather than that of the FWER is desired, there is potential for a gain in power. A simple sequential Bonferroni-type procedure is proved to control the false discovery rate for independent test statistics, and a simulation study shows that the gain in power is substantial. The use of the new procedure and the appropriateness of the criterion are illustrated with examples.

Keywords: BONFERRONI-TYPE PROCEDURES; FAMILYWISE ERROR RATE; MULTIPLE-COMPARISON PROCEDURES; *p*-VALUES

1. INTRODUCTION

When pursuing multiple inferences, researchers tend to select the (statistically) significant ones for emphasis, discussion and support of conclusions. An unguarded use of single-inference procedures results in a greatly increased false positive (significance) rate. To control this multiplicity (selection) effect, classical multiple-comparison procedures (MCPs) aim to control the probability of committing any type I error in families of comparisons under simultaneous consideration. The control of this familywise error rate (FWER) is usually required in a strong sense, i.e. under all configurations of the true and false hypotheses tested (see for example Hochberg and Tamhane (1987)).

Even though MCPs have been in use since the early 1950s, and in spite of the advocacy for their use (e.g. being mandatory for some journals, as well as in some institutions such as the Food and Drug Administration of the USA), researchers have not yet widely adopted these procedures. In medical research, for example, Godfrey (1985), Pocock *et al.* (1987) and Smith *et al.* (1987) examined samples of reports of comparative studies from major medical journals. They found that researchers overlook various kinds of multiplicity, and as a result reporting tends to exaggerate treatment differences (Pocock *et al.*, 1987).

Some of the difficulties with classical MCPs which cause their underutilization in applied research are as follows.

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- (a) Much of the methodology of FWER controlling MCPs concerns comparisons of multiple treatments and families whose test statistics are multivariate normal (or t). In practice, many of the problems encountered are not of the multiple-treatments type, and test statistics are not multivariate normal. In fact, families are often combined with statistics of different types.
- (b) Classical procedures that control the FWER in the strong sense, at levels conventional in single-comparison problems, tend to have substantially less power than the per comparison procedure of the same levels.
- (c) Often the control of the FWER is not quite needed. The control of the FWER is important when a conclusion from the various individual inferences is likely to be erroneous when at least one of them is. This may be the case, for example, when several new treatments are competing against a standard, and a single treatment is chosen from the set of treatments which are declared significantly better than the standard. However, a treatment group and a control group are often compared by testing various aspects of the effect (different end points in clinical trials terminology). The overall conclusion that the treatment is superior need not be erroneous even if some of the null hypotheses are falsely rejected.

The first difficulty has been partially addressed by the recent line of research advancing Bonferroni-type procedures, which use the observed individual p -values, while remaining faithful to FWER control: Simes (1986), Hommel (1988), Hochberg (1988) and Rom (1990). The other two difficulties still present a serious problem. This is probably why a per comparison error rate (PCER) approach, which amounts to ignoring the multiplicity problem altogether, is still recommended by some (e.g. Saville (1990)).

In this work we suggest a new point of view on the problem of multiplicity. In many multiplicity problems the number of erroneous rejections should be taken into account and not only the question whether any error was made. Yet, at the same time, the seriousness of the loss incurred by erroneous rejections is inversely related to the number of hypotheses rejected. From this point of view, a desirable error rate to control may be the expected proportion of errors among the rejected hypotheses, which we term the false discovery rate (FDR). This criterion integrates Spjøtvoll's (1972) concern about the number of errors committed in multiple-comparison problems, with Sorić's (1989) concern about the probability of a false rejection given a rejection. We use the term FDR after Sorić (1989), who identified a rejected hypothesis with a 'statistical discovery'.

After some preliminaries, we present in Section 2.1 a formal definition of the FDR. Two immediate but important consequences of controlling this error rate are given: it implies weak control of FWER and it admits more powerful procedures. In Section 2.2 we present some examples where the control of the FDR is desirable. In Section 3 we present a simple Bonferroni-type FDR controlling procedure and the rest of the section is devoted to a discussion and demonstration of its properties. Section 4 presents a simulation study of the power of the procedure.

2. FALSE DISCOVERY RATE

Consider the problem of testing simultaneously m (null) hypotheses, of which m_0 are true. $\mathbf{R}$ is the number of hypotheses rejected. Table 1 summarizes the

TABLE 1
Number of errors committed when testing m null hypotheses

	Declared non-significant	Declared significant	Total
True null hypotheses	U	V	m_0
Non-true null hypotheses	T	S	$m - m_0$
	$m - R$	R	m

situation in a traditional form. The specific m hypotheses are assumed to be known in advance. R is an observable random variable; U , V , S and T are unobservable random variables. If each individual null hypothesis is tested separately at level α , then $R = R(\alpha)$ is increasing in α . We use the equivalent lower case letters for their realized values.

In terms of these random variables, the PCER is $E(V/m)$ and the FWER is $P(V \geq 1)$. Testing individually each hypothesis at level α guarantees that $E(V/m) \leq \alpha$. Testing individually each hypothesis at level α/m guarantees that $P(V \geq 1) \leq \alpha$.

2.1. Definition of False Discovery Rate

The proportion of errors committed by falsely rejecting null hypotheses can be viewed through the random variable $Q = V/(V + S)$ —the proportion of the rejected null hypotheses which are erroneously rejected. Naturally, we define $Q = 0$ when $V + S = 0$, as no error of false rejection can be committed. Q is an unobserved (unknown) random variable, as we do not know v or s , and thus $q = v/(v + s)$, even after experimentation and data analysis. We define the FDR Q_e to be the expectation of Q ,

$$Q_e = E(Q) = E\{V/(V + S)\} = E(V/R).$$

Two properties of this error rate are easily shown, yet are very important.

- If all null hypotheses are true, the FDR is equivalent to the FWER: in this case $s = 0$ and $v = r$, so if $v = 0$ then $Q = 0$, and if $v > 0$ then $Q = 1$, leading to $P(V \geq 1) = E(Q) = Q_e$. Therefore control of the FDR implies control of the FWER in the weak sense.
- When $m_0 < m$, the FDR is smaller than or equal to the FWER: in this case, if $v > 0$ then $v/r \leq 1$, leading to $\chi_{(V \geq 1)} \geq Q$. Taking expectations on both sides we obtain $P(V \geq 1) \geq Q_e$, and the two can be quite different. As a result, any procedure that controls the FWER also controls the FDR. However, if a procedure controls the FDR only, it can be less stringent, and a gain in power may be expected. In particular, the larger the number of the non-true null hypotheses is, the larger S tends to be, and so is the difference between the error rates. As a result, the potential for increase in power is larger when more of the hypotheses are non-true.